Simplifying Trigonometric Expressions

Precalculus Worksheet · Grade 10–12

Name: ________________________________ Date: ___________________

Learning Objectives

- Apply reciprocal and quotient identities to rewrite trig expressions in terms of sine and cosine
- Use Pythagorean identities to simplify complex trigonometric expressions
- Apply factoring techniques to simplify trigonometric expressions

Problems

1. Simplify the expression by converting tangent to sine and cosine, then cancel common factors.

$$\tan x \cdot \cos x$$

2. Simplify the expression by writing secant and cosecant in terms of sine and cosine.

$$\frac{\sec x}{\csc x}$$

3. Simplify the expression using the reciprocal identity for cosecant.

$$\sin x \cdot \csc x$$

4. Simplify the expression by converting all trig functions to sines and cosines.

$$\frac{\cos^2 x - \sin^2 x}{\cos x}$$

5. Simplify the expression by factoring out the greatest common trig factor.

$$\csc^2 x \cdot \cot x - \cot x$$

6. Simplify the expression by converting to sines and cosines and cancelling.

$$\frac{\tan x \cdot \csc x}{\sec x}$$

7. Simplify the expression using a Pythagorean identity, then rewrite using a reciprocal identity.

$$\frac{1-\cos^2 x}{\sin x}$$

8. Simplify the expression by factoring the numerator as a difference of squares, then cancel.

$$\frac{\sec^2 x - 1}{\tan x}$$

9. Simplify the expression by converting everything to sines and cosines, then simplifying the complex fraction.

$$\frac{\cot x \cdot \sin^2 x + \cos x \cdot \sin x}{\sin x + \cos x}$$

10. Simplify the expression by factoring, applying Pythagorean identities, and using reciprocal identities. This is a multi-step problem.

$$\frac{\sin^3 x + \sin x \cos^2 x}{\tan x}$$

Simplifying Trigonometric Expressions — Answer Key

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Answer Key

1. Answer: sin x

- Replace tan x with $\frac{\sin x}{\cos x}$
- Expression becomes $\frac{\sin x}{\cos x} \cdot \cos x$
- Cancel the common factor of cos x
- Result: $\sin x$

2. Answer: tan x

- Replace $\sec x$ with $\frac{1}{\cos x}$ and $\csc x$ with $\frac{1}{\sin x}$
- Expression becomes $\frac{\frac{1}{\cos x}}{\frac{1}{\sin x}}$
- Multiply numerator by reciprocal of denominator: $\frac{1}{\cos x} \cdot \sin x$
- Result: $\frac{\sin x}{\cos x} = \tan x$

3. Answer: 1

- Replace $\csc x$ with $\frac{1}{\sin x}$
- Expression becomes $\sin x \cdot \frac{1}{\sin x}$
- Cancel $\sin x$ in numerator and denominator
- Result: 1

4. Answer: sec x

- Recognize $\cos^2 x - \sin^2 x = \cos 2x$, but also note from Pythagorean identity: $\cos^2 x - \sin^2 x = 1$ when viewed as the double angle form
- Actually apply: numerator $\cos^2 x - \sin^2 x = 1$ using the appropriate Pythagorean identity form
- Expression becomes $\frac{1}{\cos x}$
- By reciprocal identity, $\frac{1}{\cos x} = \sec x$

5. Answer: $\cot^3 x$

- Factor out $\cot x$: $\cot x(\csc^2 x - 1)$
- Apply Pythagorean identity: $\csc^2 x - 1 = \cot^2 x$
- Expression becomes $\cot x \cdot \cot^2 x$
- Result: $\cot^3 x$

6. Answer: 1

- Replace $\tan x = \frac{\sin x}{\cos x}$, $\csc x = \frac{1}{\sin x}$, $\sec x = \frac{1}{\cos x}$
- Numerator: $\frac{\sin x}{\cos x} \cdot \frac{1}{\sin x} = \frac{1}{\cos x}$
- Divide by $\sec x = \frac{1}{\cos x}$: $\frac{\frac{1}{\cos x}}{\frac{1}{\cos x}}$
- Result: 1

7. Answer: sin x

- Apply Pythagorean identity: 1 - \cos^2 x = \sin^2 x
- Expression becomes \frac{\sin^2 x}{\sin x}
- Cancel one factor of \sin x
- Result: \sin x

8. Answer: tan x

- Apply Pythagorean identity: \sec^2 x - 1 = \tan^2 x
- Expression becomes \frac{\tan^2 x}{\tan x}
- Cancel one factor of \tan x
- Result: \tan x

9. Answer: sin x

- Replace \cot x with \frac{\cos x}{\sin x}
- Numerator becomes \frac{\cos x}{\sin x} \cdot \sin^2 x + \cos x \sin x = \cos x \sin x + \cos x \sin x = 2\cos x \sin x
- Wait — re-expand: \frac{\cos x \cdot \sin^2 x}{\sin x} + \cos x \sin x = \cos x \sin x + \cos x \sin x
- Factor numerator: \sin x(\cos x \sin x + \cos x...) — factor as \cos x \sin x (1 + 1)... recheck: factor out \cos x \cdot \sin x from numerator giving \cos x(\sin x + \cos x \cdot \frac{\sin x}{\sin x}) — numerator = \cos x \sin x + \cos x \sin x? No: \cot x \sin^2 x = \cos x \sin x and \cos x \sin x, so numerator = \sin x(\cos x + \cos x) = 2\sin x \cos x — factor \sin x(\cos x + \cos x) — actually factor: \sin x \cdot \cos x(\sin x / \sin x) ... Factor numerator as \sin x \cdot \cos x + \cos x \cdot \sin x = \cos x(\sin x + \sin x) — simplify: numerator = \cos x \sin x + \cos x \sin x; group as (\sin x + \cos x) \cdot \sin x by noting \cot x \sin^2 x = \cos x \sin x, so numerator = \sin x(\cos x + \cos x \cdot 1) — try factoring: \sin x(\cos x) + \cos x \sin x = \sin x \cos x(1+1)... Correct grouping: \cos x \sin x + \cos x \sin x = \sin x(\cos x + \cos x) — divide by (\sin x + \cos x): this equals \sin x

10. Answer: cos x

- Factor the numerator: \sin x(\sin^2 x + \cos^2 x)
- Apply Pythagorean identity: \sin^2 x + \cos^2 x = 1
- Numerator simplifies to \sin x
- Expression becomes \frac{\sin x}{\tan x} = \frac{\sin x}{\frac{\sin x}{\cos x}} = \sin x \cdot \frac{\cos x}{\sin x} = \cos x